

5.2 Entropy (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.2 Entropy (A Level Only)
Difficulty	Medium

Time allowed: 50

Score: /40

Percentage: /100

Question 1a

The spontaneity of a chemical reaction depends on the standard Gibbs free energy change, $\Delta G^\ominus$.

State and explain whether the following processes will lead to an increase or decrease in entropy.

i)

Steam condensing to water

entropy change

explanation

[1]

ii)

Zinc reacting with nitric acid

entropy change

explanation

[1]

iii)

Sodium chloride dissolving in water

entropy change

explanation

[1]

[3 marks]

Question 1b

Calcium carbonate thermally decomposes to form calcium oxide and carbon dioxide, as shown below:



The enthalpy change of the above reaction is $\Delta H^\theta = +178 \text{ kJ mol}^{-1}$

Standard entropies are shown in Table 1.1

Table 1.1

substance	$\text{CaCO}_3(\text{s})$	$\text{CaO}(\text{s})$	$\text{CO}_2(\text{g})$
$S^\theta / \text{J K}^{-1} \text{mol}^{-1}$	+92.9	39.8	+214

Calculate the free energy change, ΔG^θ , at 298 K for this process. Include a relevant sign and give your answer to **three** significant figures. Show your working.

$\Delta G^\theta = \dots\dots\dots \text{kJ mol}^{-1}$

[3 marks]

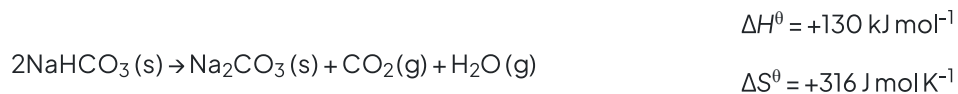
Question 1c

Explain, with reference to ΔG^θ , why this reaction becomes more feasible at higher temperatures.

[1 mark]

Question 1d

On heating, sodium hydrogencarbonate decomposes into sodium carbonate as shown.



$$\Delta H^\theta = +130 \text{ kJ mol}^{-1}$$

$$\Delta S^\theta = +316 \text{ J mol K}^{-1}$$

Calculate the minimum temperature at which the reaction becomes spontaneous. Show your working.

Temperature = K

[3 marks]

Question 2a

Ammonia, NH_3 , is produced by the Haber process and is an important chemical in the manufacture of fertilisers and clearing products.

Ammonia gas can react with oxygen to produce nitrogen monoxide and steam, and is the first step in the Ostwald process which produces nitric acid.

i)
Write an equation for the reaction of ammonia with oxygen to produce nitrogen monoxide and steam.

[2]

ii)
Table 2.1 shows the entropy values for the species involved in the reaction.

Table 2.1

Substance	Entropy values ($\text{J K}^{-1} \text{mol}^{-1}$)
$\text{NH}_3(\text{g})$	192.8
$\text{O}_2(\text{g})$	205.2
$\text{H}_2\text{O}(\text{g})$	188.8
$\text{NO}(\text{g})$	210.8

Use the data in Table 2.1 to determine the entropy for change for this reaction at 298 K.

$$\Delta S^\theta = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$$

[3]

[5 marks]

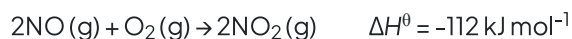
Question 2b

Explain why the standard entropy change for the reaction in part (a) is positive.

[1 mark]

Question 2c

The second step in the Ostwald process produces nitrogen dioxide as shown in the equation



The standard entropy for $\text{NO}_2(\text{g})$ is $240.0 \text{ J K}^{-1} \text{ mol}^{-1}$

Use Table 2.1 to determine the value for the free energy change, ΔG^θ , for this reaction at 298 K. Show your working.

$\Delta G^\theta = \dots\dots\dots \text{ kJ mol}^{-1}$

[6 marks]

Question 2d

Explain whether changing the temperature for the production of nitrogen dioxide will alter the spontaneity of the reaction.

[2 marks]

Question 3a

Potassium nitrate is a soluble salt. When it dissolves in water the value of the enthalpy change $\Delta H^\theta = +34.9 \text{ kJ mol}^{-1}$ and the value of the entropy change $\Delta S^\theta = +117 \text{ J K}^{-1} \text{ mol}^{-1}$.

Write an equation, including state symbols, for the process that occurs when potassium nitrate dissolves in water.

[1 mark]

Question 3b

Explain why ΔS^θ is positive for this process.

[1 mark]

Question 3c

Calculate the minimum temperature at which the reaction will become feasible. Show your working.

$T = \dots\dots\dots \text{K}$

[3 marks]

Question 3d

i)

Deduce what happens to the value of ΔG^θ when potassium nitrate dissolves in water at a temperature lower than your answer to part (c).

[1]

ii)

What does this new value of ΔG^θ suggest about the dissolving of potassium nitrate at this lower temperature?

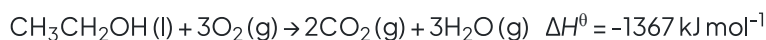
[1]

[2 marks]

Question 4a

Ethanol is used in large quantities in the production of alcoholic beverages and as a fuel.

The combustion of ethanol is represented by the equation



Using Table 4.1, determine the entropy change, ΔS^θ , for the combustion of ethanol at 298K. Show your working.

Table 4.1

substance	$\text{CH}_3\text{CH}_2\text{OH}(\text{l})$	$3\text{O}_2(\text{g})$	$2\text{CO}_2(\text{g})$	$3\text{H}_2\text{O}(\text{g})$
$S^\theta / \text{J K}^{-1} \text{mol}^{-1}$	+161	+205.2	+213.8	+188.8

$\Delta S^\theta = \dots\dots\dots (\text{J K}^{-1} \text{mol}^{-1})$

[3 marks]

Question 4b

Calculate the free energy change, $\Delta G^\ominus$, for the combustion of ethanol using $\Delta S^\ominus$ determined in part (a).

[3 marks]

Question 4c

Explain whether changing the temperature for the combustion of ethanol will alter the spontaneity of the reaction.

[3 marks]

Model Answers: Medium

1a

a)

i) Entropy:

- Decreases / is negative
AND
Decrease in gas; [1 mark]

ii) Entropy:

- Increases / is positive
AND
 H_2 / gas is formed; [1 mark]

iii) Entropy:

- Increases / is positive
AND
 NaCl(aq) solution has free moving / mobile ions / aqueous ions; [1 mark]

[Total: 3 marks]

- Entropy increases when the system has become more disordered
- It can help to write out the equations to identify the changes
 - $\text{H}_2\text{O(g)} \rightarrow \text{H}_2\text{O(l)}$ - gas turning into a liquid so order is increasing and therefore entropy decreases
 - $\text{Zn(s)} + 2\text{HNO}_3\text{(aq)} \rightarrow \text{Zn(NO}_3)_2\text{(aq)} + \text{H}_2\text{(g)}$ - gas being formed so disorder is increasing and therefore entropy increases
 - $\text{NaCl(s)} + \text{aq} \rightarrow \text{Na}^+\text{(aq)} + \text{Cl}^-\text{(aq)}$ - ions are now able to move so disorder is increasing and therefore entropy increases

1b

b) To calculate the free energy change, ΔG^θ , for this process:

- $\Delta S^\theta = (39.8 + 214) - 92.9 = (+)160.9 \text{ J K}^{-1} \text{ mol}^{-1}$; [1 mark]

- $\Delta G^\theta = 178 - (298 \times \frac{160.9}{1000})$

OR

$$\Delta G^\theta = 178000 - (298 \times 160.9); [1 \text{ mark}]$$

- $\Delta G^\theta = (+)130 \text{ (kJ mol}^{-1}\text{)}$ to **three** significant figures; [1 mark]

[Total: 3 marks]

- The two equations required are:

- $\Delta S^\theta = \sum S_{\text{products}}^\theta - \sum S_{\text{reactants}}^\theta$

- $\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$

- **Remember:** For this calculation, the units for ΔH and ΔS must match, so must be in J or kJ so:

- Either multiply $\Delta H(\text{kJ mol}^{-1})$ by 1000

OR

- Divide $\Delta S(\text{J K}^{-1} \text{ mol}^{-1})$ by 1000

- **Careful:** The final answer must be in kJ mol^{-1} , so if you had converted to J mol^{-1} , you must remember to convert back to kJ mol^{-1}

1c

c) This reaction becomes more feasible at higher temperatures because:

- $T\Delta S^\theta$ is more positive than ΔH^θ

AND

ΔG^θ is negative / decrease / less positive; [1 mark]

[Total: 1 mark]

- For a reaction to be feasible, ΔG^θ must be negative
- It is important to be able to explain the effect on the free energy of a reaction dependant on the value of ΔH^θ and ΔS^θ

If $\Delta H^\theta \dots$	And if $\Delta S^\theta \dots$	Then ΔG^θ is...	Spontaneous...
Is positive > 0 endothermic	Is positive > 0 More disorder	Negative at high T Positive at low T	Depending on T

$$\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$$

- At higher temperatures, $-T\Delta S^\theta$ will become more and more negative, and will eventually outweigh the effect of ΔH
- The reaction won't be feasible at low temperatures, but if you heat it, there will be a temperature at which it becomes feasible, because ΔG^θ becomes negative
- Such a process is spontaneous at high temperatures and non-spontaneous at low temperatures
- Thermal decomposition reactions are good examples of where both ΔH^θ and ΔS^θ are positive

1d

d) To calculate the minimum temperature at which the reaction becomes spontaneous:

- $\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta = 0$; [1 mark]
- $T = \frac{\Delta H^\theta}{\Delta S^\theta} = \frac{130}{\left(\frac{316}{1000}\right)}$; [1 mark]
- $T = 410 / 411 / 412 / 411.4 \text{ (K)}$; [1 mark]

[Total: 3 marks]

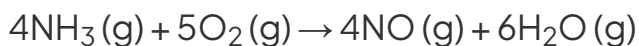
- You should be able to rearrange $\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$ in order to work out the minimum temperature at which the reaction becomes feasible
 - This will occur when $\Delta G^\theta = 0$
- **Remember:** For this calculation, the units for ΔH^θ and ΔS^θ must match, so be in J or kJ so:
 - Either multiply ΔH^θ (kJ mol⁻¹) by 1000
 - OR**
 - Divide ΔS^θ (J K⁻¹ mol⁻¹) by 1000
- Pick whichever one you find the easiest to use and remember

2a

a)

i) The equation for the reaction of ammonia with oxygen to produce nitrogen

monoxide and steam is:



- Correct reactants, products and balancing; [1 mark]
- Correct state symbols; [1 mark]

ii) To calculate the free energy change at 298 K:

- $\Delta S^\theta = \Sigma S^\theta(\text{products}) - \Sigma S^\theta(\text{reactants})$; [1 mark]
- $\Delta S^\theta = [(4 \times S^\theta\text{NO}(\text{g})) + (6 \times S^\theta\text{H}_2\text{O}(\text{g}))] - [(4 \times S^\theta\text{NH}_3(\text{g})) + (5 \times S^\theta\text{O}_2(\text{g}))]$

OR

- $\Delta S^\theta = [(4 \times 210.8) + (6 \times 188.8)] - [(4 \times 192.8) + (5 \times 205.2)]$; [1 mark]
- $\Delta S^\theta = (843.2 + 1132.8) - (771.2 + 1026)$
- $\Delta S^\theta = 1976 - 1797.2$
- $\Delta S^\theta = (+)178.8 \text{ (J K}^{-1} \text{ mol}^{-1})$; [1 mark]

[Total: 5 marks]

- This is a tricky equation to balance so take your time
- This balancing must be transferred into the calculation to determine the entropy change for the reaction
- Don't forget to take into account the number of moles of each reactant and product when calculating entropy and free energy from ΔS^θ and ΔG^θ values respectively

2b

b) The standard entropy change for the reaction outlined in part (a) is positive because

- There are 10 moles of gas in the products and 9 moles of gas in the reactants
- OR**
- There are more moles of gas in the products than the reactants; [1 mark]

[Total: 1 mark]

- Make sure you make reference to the moles of gas each side of the reaction

2c

c) To determine the value for the free energy change for this reaction:

Determine ΔS^θ

- $\Delta S^\theta = \Sigma S(\text{products}) - \Sigma S(\text{reactants})$

OR

$$\Delta S^\theta = [(2 \times \Delta S^\theta \text{NO}_2(\text{g}))] - [(2 \times \Delta S^\theta \text{NO}(\text{g})) + (\Delta S^\theta \text{O}_2(\text{g}))]$$

OR

$$\Delta S^\theta = (2 \times 240.0) - (2 \times 210.8) + (205.2); [1 \text{ mark}]$$

- $\Delta S^\theta = 480 - 626.8; [1 \text{ mark}]$
- $\Delta S^\theta = -146.8 (\text{J K}^{-1} \text{mol}^{-1}); [1 \text{ mark}]$

Determine ΔG^θ

- $\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta; [1]$
- $\Delta G^\theta = -112 - 298(-0.1468); [1]$
- $\Delta G^\theta = -68.3 (\text{kJ mol}^{-1}); [1]$

[Total: 6 marks]

- As you are given ΔS^θ for the reaction, this should be determined first and then substitute this value into $\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$
- Units must be taken into account so be careful to divide the value for ΔS^θ by 1000

2d

d) Explaining whether changing the temperature alters the spontaneity of the reaction:

- Changing the temperature will alter the spontaneity of the reaction

OR

The spontaneity of the reaction is dependent on T; [1 mark]

- Decreasing the temperature will reduce the value of free energy / ΔG^θ / free energy / ΔG^θ becomes more negative

OR

Increasing the temperature will increase the value of free energy / ΔG^θ / free energy / ΔG^θ becomes more positive; [1 mark]

[Total: 2 marks]

- It is important to be able to explain the effect on the free energy of a reaction dependant on the value of ΔH^θ and ΔS^θ

If ΔH^θ ...	And if ΔS^θ ...	Then ΔG^θ is...	Spontaneous...
Is negative < 0 exothermic	Is negative < 0 More order	Negative at low T Positive at high T	Depending on T

$$\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$$

- As temperature increases, $-T\Delta S^\theta$ will become more and more positive, and will eventually outweigh the effect of ΔH^θ
- The value of ΔG^θ will then become positive, and the reaction will no longer be feasible
- Such a process is spontaneous at low temperatures and non spontaneous at high temperatures

3a

a) An equation, including state symbols, for the process that occurs when potassium nitrate dissolves in water is:

- $\text{KBr (s)} \rightarrow \text{K}^+ \text{(aq)} + \text{NO}_3^- \text{(aq)}$

OR

$\text{KNO}_3 \text{(s)} \rightarrow \text{K}^+ \text{(aq)} + \text{NO}_3^- \text{(aq)}$

energy / ΔG^θ becomes more negative

OR

Increasing the temperature will increase the value of free energy / ΔG^θ / free energy / ΔG^θ becomes more positive; [1 mark]

[Total: 2 marks]

- It is important to be able to explain the effect on the free energy of a reaction dependant on the value of ΔH^θ and ΔS^θ

If ΔH^θ	And if ΔS^θ	Then ΔG^θ is....	Spontaneous....
Is negative < 0 exothermic	Is negative < 0 More order	Negative at low T Positive at high T	Depending on T

$$\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$$

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- The value of ΔG^θ will then become positive, and the reaction will no longer be feasible
- Such a process is spontaneous at low temperatures and non spontaneous at high temperatures

3a

a) An equation, including state symbols, for the process that occurs when potassium nitrate dissolves in water is:

- $\text{KBr (s)} \rightarrow \text{K}^+ \text{(aq)} + \text{NO}_3^- \text{(aq)}$

OR

$\text{KBr (s)} + \text{aq} \rightarrow \text{K}^+ \text{(aq)} + \text{NO}_3^- \text{(aq)}$; [1 mark]

[Total: 1 mark]

- The standard enthalpy change of solution ($\Delta H_{\text{sol}}^\theta$) is the enthalpy change when 1 mole of an ionic substance dissolves in sufficient water to form a very dilute solution

- The symbol (aq) is used to show that the solid is dissolved in sufficient water

3b

b) The sign for $\Delta S^\ominus$ is positive because:

- Increase in disorder because solid $\rightarrow$ solution / increase in number of particles per mole of solution; [1 mark]

[Total: 1 mark]

- $\text{KBr(s)} + \text{aq} \rightarrow \text{K}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$
 - Ions are now able to move so disorder has increased and therefore entropy increases and $\Delta S^\ominus$ is positive

3c

c) To calculate the minimum temperature at which the reaction becomes feasible:

- $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus = 0$; [1 mark]
- $T = \frac{\Delta H^\ominus}{\Delta S^\ominus} = \frac{34.9}{\left(\frac{117}{1000}\right)}$; [1 mark]
- $T = 298 \text{ (K)}$; [1 mark]

[Total: 3 marks]

- You should be able to rearrange $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$ in order to work out the minimum temperature at which the reaction becomes feasible
 - This will occur when $\Delta G^\ominus = 0$
- **Remember:** For this calculation, the units for $\Delta H^\ominus$ and $\Delta S^\ominus$ must match, so be in J or kJ so:
 - Either multiply $\Delta H^\ominus$ (kJ mol⁻¹) by 1000
 - OR**
 - Divide $\Delta S^\ominus$ (J K⁻¹ mol⁻¹) by 1000
- Pick whichever one you find the easiest to use and remember

3d

d)

i) The value of ΔG^θ becomes:

- Positive / increases / $\Delta G^\theta > 0$ Allow more positive; [1 mark]

ii) The new value of ΔG^θ suggests that:

- If (d)(i) positive, dissolving is no longer spontaneous / no longer feasible / potassium nitrate does not dissolve / less

OR

If (d)(i) negative, dissolving is spontaneous / feasible / potassium nitrate dissolves / more soluble; [1 mark]

[Total: 2 marks]

- For this process, remember you are looking at the signs of the enthalpy and entropy change in order to determine if the reaction is feasible;
 - In this case you have a **positive ΔH^θ** and a **positive ΔS^θ**
 - This means that the **ΔG^θ will be positive depending on T**

If ΔH^θ ...	And if ΔS^θ ...	Then ΔG^θ is ...	Spontaneous ...
Is positive > 0 endothermic	Is positive > 0 More disorder	Negative at high T Positive at low T	Depending on T

$$\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$$

- At higher temperatures, $-T\Delta S^\theta$ will become more and more negative, and will eventually outweigh the effect of ΔH^θ
- The reaction won't be feasible at low temperatures, but if you heat it, there will be a temperature at which it becomes feasible, because ΔG^θ becomes negative
- Such a process is spontaneous at high temperatures and non-spontaneous at low temperatures

4a

a) To determine the entropy change, ΔS^θ , for the combustion of ethanol at 298 K:

- $\Delta S^\theta = \Sigma S^\theta(\text{products}) - \Sigma S^\theta(\text{reactants})$

$$\Delta S^\theta = [(2 \times 213.8) + (3 \times 188.8)] - [(1 \times 161) + (3 \times 205.2)]; [1 \text{ mark}]$$

- $\Delta S^\theta = (427.6 + 566.4) - (161 + 615.6)$

OR

$$\Delta S^\theta = 994 - 776.6; [1 \text{ mark}]$$

- $\Delta S^\theta = (+)217.4 \text{ (J K}^{-1} \text{ mol}^{-1}); [1 \text{ mark}]$

[Total: 3 marks]

- Careful: You must multiply the ΔS^θ values depending on their stoichiometry in the equation, so it should be
 - $3 \times \Delta S^\theta \text{H}_2\text{O (g)}$
 - $2 \times \Delta S^\theta \text{CO}_2$
 - $3 \times \Delta S^\theta \text{O}_2$

4b

b) To calculate the standard free energy for the combustion of ethanol:

- $\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta; [1 \text{ mark}]$
- $\Delta G^\theta = -1367 - (298 \times 0.2174); [1 \text{ mark}]$
- $\Delta G^\theta = -1432 \text{ (kJ mol}^{-1}); [1 \text{ mark}]$

[Total: 3 marks]

- For free-energy change calculations, the value and units of ΔS^θ must be changed to $\text{kJ}^{-1} \text{ mol}^{-1}$ by dividing by 1000
 - This is where the (0.2174) comes from in the equation; $\frac{-217.4}{1000}$
 - This then matches the kJ in ΔH^θ
 - Without this conversion, your answer will be wrong and could then effect any further calculations
- It is possible to change the units for ΔH^θ instead from kJ mol^{-1} to J mol^{-1} , by multiplying by 1000, though it is best to stick with one method as otherwise it can cause confusion
- Examiners commented on issues with units being widespread so take care to practise this!

4c

c) Explaining whether changing the temperature alters the spontaneity for the

combustion of ethanol:

- Not alter the spontaneity of the reaction; [1 mark]
- (Because) ΔH^θ for this reaction is always negative
AND
 ΔS^θ is always positive for this reaction; [1 mark]
- Therefore ΔG^θ will always be negative at any temperature and therefore feasible / spontaneous; [1 mark]

[Total: 3 marks]

- It is important to be able to explain the effect on the free energy of a reaction dependant on the value of ΔH^θ and ΔS^θ

If ΔH^θ ...	And if ΔS^θ ...	Then ΔG^θ is...	Spontaneous...
Is negative < 0 exothermic	Is positive > 0 More disorder	Always negative < 0	Always

- In this case, ΔG will be negative regardless of the temperature
- Such a process is feasible at all temperatures

